Lyapunov energies for convergence and alternation in IGAHD

Abstract

We give explicit Lyapunov certificates for four regimes of the original IGAHD recurrence. In the scalar case with $\alpha = 4$ and $s = 1$, a phase energy at $\beta = 9/4$ proves global convergence and finite length for C^2 potentials with $1/100 \le f'' \le 1$. For coercive C^2 potentials with $0 \le f'' \le 1$ and minimizer interval $[-1, 1]$, two weighted endpoint energies prove point convergence without a positive curvature lower bound. At $\beta = 3$, a fixed smooth strongly convex potential has an open basin of nonminimizing alternating limits. A separate result for C^4 objectives treats a compact Morse–Bott minimum manifold, fixed $\alpha > 3$, $\beta = \sqrt{s}$ and $0 < sL \le 1/4$. A moving normal-history metric and a clock-weighted Lyapunov function capture all sufficiently close independent starting pairs and give finite travel to an unspecified minimum. All four regimes retain the actual predictors, old-gradient correction and finite negative-inertia prefix. Classical interpolation, momentum, perturbation and invariant-manifold ideas are credited. The conclusions concern these explicit certificates and their stated domains, without a sharp parameter threshold or a comprehensive novelty claim.

Keywords: inertial dynamics; Hessian-driven damping; Lyapunov energy; nonautonomous recurrence; omega-limit set

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The first seven sections retain three scalar regimes with $\alpha = 4$ and $s = 1$. Section 8 treats a separate local, finite-dimensional regime with $\beta = \sqrt{s}$ and a structurally small fixed step. The latter is not an extension of the scalar parameter range, and finite length is not asserted for the flat-interval class.

1 The trajectory and its phase energy

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable and put $g = f'$. The inertial gradient algorithm with Hessian-driven damping, abbreviated IGAHD, is

$$\begin{aligned} y_k &= x_k + \left(1 - \frac{\alpha}{k}\right)(x_k - x_{k-1}) - \beta\sqrt{s}\,[g(x_k) - g(x_{k-1})] - \frac{\beta\sqrt{s}}{k} g(x_{k-1}), \\ x_{k+1} &= y_k - s g(y_k), \qquad k \ge 1, \end{aligned} \tag{1}$$

with two arbitrary original initial points $x_0, x_1 \in \mathbb{R}$. This is the recurrence introduced by Attouch, Chbani, Fadili and Riahi [1], with iterate convergence studied in [2]. In particular, the final correction in the first line is evaluated at x_{k-1}.

For $\alpha \ge 3$ and convex potentials with an L-Lipschitz gradient and a minimizer, Adly [3, Theorem 2.1] gives a sufficient nonlinear energy condition

$$0 < \beta L\sqrt{s} < 1 + \sqrt{1 + sL(1 - sL)}, \qquad 0 < s \le 1/L. \tag{2}$$

For quadratic potentials, the same source proves convergence of the original recurrence on a larger region. At $s = L = 1$, the nonlinear sufficient condition is $\beta < 2$, whereas its uniform quadratic spectral condition is $\beta < 9/2$ [3, Propositions 6.1–6.2]. The source asks whether another Lyapunov function can recover part or all of this larger region for nonquadratic potentials. We examine the fixed point

$$\alpha = 4, \qquad \beta = 3, \qquad s = L = 1. \tag{3}$$

The result below concerns the actual trajectory at these parameters. Its central estimate is a local energy for convergence to an alternating limit.

For $k \geq 1$, define the signed phase error

$$w_k = \begin{pmatrix} (-1)^k x_k - 1 \\ (-1)^{k-1} x_{k-1} - 1 \end{pmatrix}. \tag{4}$$

We use the following fixed matrix and constants:

$$P = \begin{pmatrix} \dfrac{988750}{144419} & \dfrac{859905}{288838} \\ \dfrac{859905}{288838} & \dfrac{95866471}{28883800} \end{pmatrix}, \qquad T = \operatorname{tr} P = \frac{293616471}{28883800}, \tag{5}$$

$$\begin{aligned} &R = \frac{1}{20}, \qquad N = \left\lceil \frac{1272}{5} T^2 \right\rceil = 26289, \qquad r = 1 - \frac{1}{4T}, \qquad D = \frac{159}{50} T, \\ &\eta = \frac{1 + r^2}{2}, \qquad C = \frac{D^2(1 + r^2)}{1 - r^2}. \end{aligned} \tag{6}$$

The norm induced by P will be denoted by $\|w\|_P = (w^{\mathsf{T}} P w)^{1/2}$.

Theorem 1. *There is one fixed even C^∞ potential f, defined in* (13) *below, such that*

$$f(0) = 0, \qquad \frac{1}{100} \leq f''(x) \leq 1 \quad (x \in \mathbb{R}), \qquad \sup_{x \in \mathbb{R}} f''(x) = 1. \tag{7}$$

Its unique minimizer is 0. *For the original recurrence* (1) *at* (3)*, the following assertions hold.*

1. *$P \geq I$ and $0 < r, \eta < 1$. Every actual trajectory with $\|w_N\|_P < R$ remains in that ball for all original $k \geq N$ and satisfies*

$$\|w_{k+1}\|_P \leq r \|w_k\|_P + \frac{D}{k}. \tag{8}$$

Its energies

$$E_k = w_k^{\mathsf{T}} P w_k, \qquad V_k = E_k + C \sum_{j=k}^{\infty} \frac{1}{j^2} \tag{9}$$

obey

$$V_k \geq E_k \geq \|w_k\|^2 \geq 0, \qquad V_{k+1} - V_k \leq -(1 - \eta) E_k. \tag{10}$$

Consequently, $\sum_{k \geq N} E_k < \infty$ and $w_k \to 0$.

2. *There is a nonempty open set $U \subset \mathbb{R}^2$ of original pairs (x_0, x_1) whose actual state at index N satisfies this strict ball condition. For every pair in U, the entire original orbit is bounded and*

$$\begin{aligned} &x_{2j} \longrightarrow 1, \qquad x_{2j+1} \longrightarrow -1, \\ &\omega(x_k) = \{-1, 1\}, \qquad \omega((x_k, x_{k-1})) = \{(1,-1),(-1,1)\}, \\ &|g(x_k)| \longrightarrow \frac{9}{10}, \qquad |x_k - x_{k-1}| \longrightarrow 2, \qquad f(x_k) \longrightarrow f(1) \geq \frac{1}{200} > 0. \end{aligned} \tag{11}$$

Here ω denotes the set of subsequential limits.

3. *At the same parameters, every scalar quadratic $f_h(x) = \frac{h}{2}(x - x^*)^2 + f^*$ with $h \in [1/100, 1]$ has $x_k \to x^*$ from every original starting pair.*

The correction in (9) is the tail of a summable series and tends to zero. Its sequence of tails is not summable over k, and no such assertion is used. The energy measures distance from the alternating phase, rather than from the minimizer. In particular, the theorem does not assert exact periodicity of the original nonautonomous trajectory.

Section 5 gives a complementary phase energy at $\beta = 9/4$, with the same $\alpha = 4$ and $s = 1$. It proves convergence from every original pair for every C^2 potential with $1/100 \leq f'' \leq 1$. The two energies therefore describe different actual limits within the same curvature class. The source comparison for the positive result retains the unresolved related-work qualification stated in Section 7.

The separation between quadratic stability and nonlinear cycling is classical. Lessard, Recht and Packard [4, Section 4.6 and Appendix B] give a scalar strongly convex heavy-ball example with an attracting cycle. Goujaud, Taylor and Dieuleveut [5, Sections 4–5] establish attraction, trapping under small time-varying heavy-ball perturbations, and smooth regularization that preserves locally affine gradients. Goujaud, Dieuleveut and Taylor [6] give a constructive framework for stationary methods. We use the same elementary stability and smoothing principles. The point here is their realization for (1), including all three gradient locations, its $1/k$ terms and an admissible original-index basin. None of the cited cycle results supplies that compatibility directly.

2 A fixed potential and its local recurrence

Put $a = 89/99$ and $b = 15/13$. Define a continuous odd function g_0 by

$$g_0(x) = \begin{cases} x, & 0 \leq x \leq a, \\ x/100 + 89/100, & a \leq x \leq b, \\ 2x/5 + 11/25, & x \geq b, \end{cases} \qquad g_0(-x) = -g_0(x). \tag{12}$$

Let $\rho(t) = c_0 \exp[-1/(1-t^2)]$ for $|t| < 1$ and $\rho(t) = 0$ otherwise, where $c_0 > 0$ is chosen so that $\int_{\mathbb{R}} \rho = 1$. Set

$$\epsilon = \frac{1}{100}, \qquad \rho_\epsilon(t) = \epsilon^{-1}\rho(t/\epsilon), \qquad g = g_0 * \rho_\epsilon, \qquad f(x) = \int_0^x g(u)\,du. \tag{13}$$

This definition is independent of the step index, the initialization and any terminal horizon.

Lemma 1. *The potential* (13) *has all the properties in* (7)*, and* $f(x) \geq x^2/200$ *for every* x*. Moreover,*

$$g(x) = \frac{x}{100} + \frac{89}{100}\operatorname{sign}(x) \qquad \text{if } \big||x| - 1\big| \leq R, \tag{14}$$

$$g(x) = \frac{2x}{5} + \frac{11}{25}\operatorname{sign}(x) \qquad \text{if } |x| \geq b + \epsilon. \tag{15}$$

In particular, $g(1) = 9/10$ *and* $g(12/5) = 7/5$*.*

Proof. The formulas in (12) agree at a and b. Its odd extension is locally absolutely continuous, with almost-everywhere derivative between $1/100$ and 1. The nonnegative smooth mollifier is even, has compact support and mass one. Thus

$$g'(x) = \int_{\mathbb{R}} g_0'(x-t)\rho_\epsilon(t)\,dt \in [1/100, 1].$$

Convolution gives $g \in C^\infty$, and evenness of the kernel preserves oddness of g_0. Hence f is even and smooth, and $g(0) = 0$. Integrating the strong monotonicity inequality yields $f(x) \geq x^2/200$. This also proves coercivity and the uniqueness of its minimizer.

The kernel has first moment zero, so convolution preserves an affine branch whenever its sampling interval stays within that branch. On $|x| \leq a - \epsilon$, it gives $g(x) = x$. Therefore the upper curvature bound is attained. For $19/20 \leq x \leq 21/20$, the sampling interval stays strictly between a and b, since

$$\frac{19}{20} - \epsilon - a = \frac{203}{4950} > 0, \qquad b - \left(\frac{21}{20} + \epsilon\right) = \frac{61}{650} > 0.$$

This proves (14) on the positive side, and oddness gives the negative side. The same argument on the outer branches proves (15), including its boundary. The two stated gradient values follow by substitution. ∎

Lemma 2. *Suppose* $k \geq 5$ *and both coordinates of* w_k *have absolute value at most* R*. Then the actual intermediate point* y_k *is in the opposite-sign outer region of* (15)*, and*

$$w_{k+1} = \left(A + \frac{B}{k}\right) w_k + \frac{d}{k}, \tag{16}$$

where

$$A = \begin{pmatrix} -591/500 & -291/500 \\ 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 12/5 & 1191/500 \\ 0 & 0 \end{pmatrix}, \qquad d = \begin{pmatrix} 159/50 \\ 0 \end{pmatrix}. \tag{17}$$

Proof. Write $\sigma = (-1)^k$ and $w_k = (p, q)^{\mathsf{T}}$. Then $x_k = \sigma(1 + p)$ and $x_{k-1} = -\sigma(1 + q)$. The input gradients are given by (14). Substitution into the literal first line of (1) gives

$$\sigma y_k = -\frac{12}{5} - \frac{53}{10k} + \left(\frac{197}{100} - \frac{4}{k}\right) p + \left(\frac{97}{100} - \frac{397}{100k}\right) q. \tag{18}$$

For $k \geq 5$, both displayed coefficients of p, q are nonnegative and their sum is at most $147/50$. Consequently,

$$\sigma y_k \leq -\frac{12}{5} + \frac{147}{50} R = -\frac{2253}{1000} < -(b + \epsilon). \tag{19}$$

The positive gap in the last inequality is $14159/13000$. Thus the output gradient may now be evaluated by (15), giving

$$x_{k+1} = \frac{3}{5} y_k + \frac{11}{25} \sigma.$$

Taking the next signed errors proves (16)–(17). In particular, the forcing d/k and the entry $1191/(500k)$ are retained. ∎

3 Coercivity, trapping and descent

We next prove the energy assertions of Theorem 1. Every use of the local recurrence will be justified by the invariant-region induction.

Exact multiplication gives

$$A^{\mathsf{T}} P A - P = -I, \tag{20}$$

and the principal minors of $P - I$ satisfy

$$P_{11} - 1 = \frac{844331}{144419} > 0, \qquad \det(P - I) = \frac{28341121761}{6036714200} > 0. \tag{21}$$

Thus $P \geq I$. Since a positive matrix has largest eigenvalue at most its trace,

$$I \leq P \leq TI, \qquad \|v\| \leq \|v\|_P \leq \sqrt{T}\|v\|. \tag{22}$$

The identity (20) consequently gives

$$\|Av\|_P^2 = \|v\|_P^2 - \|v\|^2 \leq \left(1 - \frac{1}{T}\right) \|v\|_P^2, \qquad \|A\|_P \leq 1 - \frac{1}{2T}. \tag{23}$$

The last inequality follows from $\sqrt{1 - t} \leq 1 - t/2$ for $0 \leq t \leq 1$.

The single nonzero row of B has squared Euclidean norm $2858481/250000 < 16$. Since $T > 1$, (22) gives

$$\|Bv\|_P \leq 4T\|v\|_P, \qquad \|d\|_P \leq \frac{159}{50}\sqrt{T} \leq D. \tag{24}$$

All constants in (6) suffice as written. Indeed,

$$\frac{1272}{5} T^2 = \frac{13707490494756520719}{521421189025000} \in (26288, 26289), \tag{25}$$

and hence

$$N \geq \frac{1272}{5}T^2 = \frac{4TD}{R} > 16T^2, \qquad \frac{D}{N} \leq \frac{R}{4T} = (1-r)R. \tag{26}$$

Combining (23) and (24), we obtain for $k \geq N$

$$\left\| A + \frac{B}{k} \right\|_P \leq 1 - \frac{1}{2T} + \frac{4T}{k} \leq 1 - \frac{1}{4T} = r < 1. \tag{27}$$

Now suppose that the actual trajectory has $\|w_N\|_P < R$. At index N, (22) places both signed input errors in $(-R, R)$. Lemma 2 therefore applies at this step and first verifies the intermediate-point region. It then yields

$$\|w_{N+1}\|_P \leq r\|w_N\|_P + D/N < rR + (1-r)R = R.$$

The same implication holds at every subsequent original index: if $\|w_k\|_P < R$, the two input gradients are in their middle pieces, (19) places y_k in its outer piece, and (16) together with (26)–(27) gives

$$\|w_{k+1}\|_P \leq r\|w_k\|_P + D/k < R.$$

Induction proves the strict invariant ball and (8) for all $k \geq N$. In particular, the proof has not assumed that the future trajectory remains in the local affine regions.

It remains to obtain a decreasing energy from this estimate. Young's inequality gives, for $z \geq 0$,

$$2rz\frac{D}{k} \leq \frac{1-r^2}{2}z^2 + \frac{2r^2D^2}{(1-r^2)k^2}.$$

Squaring (8) and applying this inequality with $z = \|w_k\|_P$ yields

$$E_{k+1} \leq \eta E_k + \frac{C}{k^2}. \tag{28}$$

The correction $S_k = \sum_{j=k}^{\infty} j^{-2}$ is finite and satisfies $S_{k+1} - S_k = -k^{-2}$. Thus $V_k = E_k + CS_k$ satisfies precisely (10). For every $M \geq N$,

$$(1-\eta)\sum_{k=N}^{M} E_k \leq V_N - V_{M+1} \leq V_N. \tag{29}$$

It follows that $E_k \to 0$, $w_k \to 0$ and $V_k \to 0$. This proves the first part of Theorem 1. The nonsummable $1/k$ forcing in the literal recurrence has been accounted for by its square in (28); it was never discarded.

4 The basin at the original initial index

The preceding argument starts from a condition on the actual state at index N. We now prove that this condition has a nonempty open preimage among the allowed original

pairs. This step must retain the entire prefix of (1), whose first inertia coefficients are $-3, -1, -1/3, 0$.

Define

$$\begin{aligned} H(t) &= t - g(t), \\ \Phi_k(t) &= -\left(1 - \frac{4}{k}\right) t + 3\left(1 - \frac{1}{k}\right) g(t), \\ F_k(u, v) &= \left(H\left((2 - 4/k)u - 3g(u) + \Phi_k(v)\right), u\right). \end{aligned} \tag{30}$$

By (1) and (3), F_k is the actual transition from (x_k, x_{k-1}) to (x_{k+1}, x_k).

Lemma 3. *For every original integer $k \geq 1$, the map $F_k : \mathbb{R}^2 \to \mathbb{R}^2$ is continuous and onto.*

Proof. Continuity follows from that of g. On the two outer branches,

$$\begin{aligned} H(t) &= \frac{3}{5} t - \frac{11}{25} \operatorname{sign}(t), \\ \Phi_k(t) &= \frac{k + 14}{5k} t + \frac{33}{25}\left(1 - \frac{1}{k}\right) \operatorname{sign}(t). \end{aligned} \tag{31}$$

Each scalar map has opposite infinite limits at the two ends of the line. The coefficient $(k+14)/(5k)$ is positive at every original index, including $k = 1, 2, 3, 4$. The intermediate value theorem makes both maps onto.

For any prescribed output (z, u), choose y with $H(y) = z$. Next choose v with

$$\Phi_k(v) = y - (2 - 4/k)u + 3g(u).$$

Then $F_k(u, v) = (z, u)$. This uses only surjectivity. In fact, H is zero on $[-a + \epsilon, a - \epsilon]$ and is not injective. ∎

Let

$$\begin{aligned} F &= F_{N-1} \circ \cdots \circ F_1, \\ e_N(u, v) &= \begin{pmatrix} (-1)^N u - 1 \\ (-1)^{N-1} v - 1 \end{pmatrix}, \\ U_N &= \{(u, v) \in \mathbb{R}^2 : e_N(u, v)^{\mathsf{T}} P e_N(u, v) < R^2\}, \\ U &= \{(x_0, x_1) \in \mathbb{R}^2 : F(x_1, x_0) \in U_N\}. \end{aligned} \tag{32}$$

The terminal ellipsoid U_N is nonempty and open. Its center is $(-1, 1)$ because N is odd. By Lemma 3, F is continuous and onto, so U is nonempty and open. The order of the composition is important: F_1 takes (x_1, x_0) to (x_2, x_1), and $F_{N-1} \circ \cdots \circ F_1$ ends at (x_N, x_{N-1}). Thus (32) specifies genuine initializations at $k = 1$, rather than introducing new data at N.

For each pair in U, Section 3 applies to its own original tail. All its earlier terms are finite because the recurrence is defined everywhere, and all tail positions have absolute

value less than $1 + R$. Hence the entire orbit is bounded. Since $w_k \to 0$, its two parity subsequences have the limits in (11). Both occur along infinite subsequences, and there are no other subsequential limits. Continuity and oddness of g give $|g(x_k)| \to 9/10$; the identity

$$x_k - x_{k-1} = (-1)^k\big(2 + (w_k)_1 + (w_k)_2\big)$$

gives the velocity limit. Finally, evenness and continuity of f give $f(x_k) \to f(1) \geq 1/200$. This proves the second part of Theorem 1.

The result asserts an open set of unrestricted two-point starts. It does not assert failure from an equal-history start $x_0 = x_1$, or from every starting pair. Nor is $w_k = 0$ an exact original trajectory: (16) would then give $w_{k+1} = d/k \neq 0$.

5 A global phase energy at a second damping

In this section we fix

$$\alpha = 4, \qquad \beta = \frac{9}{4}, \qquad s = 1, \qquad \frac{1}{100} \leq f''(t) \leq 1 \quad (t \in \mathbb{R}), \tag{33}$$

where $f \in C^2(\mathbb{R})$. Let x^* be its unique minimizer, put $f^* = f(x^*)$ and $g = f'$, and define

$$\begin{aligned} p(x,z) &= x - z + \frac{9}{4}g(z), \\ W(x,z) &= f(x) - f^* + \frac{1}{2}p(x,z)^2 + \frac{1}{2}g(x)^2 + \frac{7}{64}g(z)^2, \\ R(x,z)^2 &= (x - x^*)^2 + (z - x^*)^2. \end{aligned} \tag{34}$$

For the original trajectory (1), write $W_k = W(x_k, x_{k-1})$, $R_k = R(x_k, x_{k-1})$, and set

$$\rho = 1 - \frac{7}{2560000} \in (0,1). \tag{35}$$

Theorem 2. *Under* (33)*, every original pair $x_0, x_1 \in \mathbb{R}$ defines a unique trajectory. The energy* (34) *satisfies, for all $x, z \in \mathbb{R}$,*

$$\frac{7}{640000}R(x,z)^2 \leq W(x,z) \leq 2R(x,z)^2. \tag{36}$$

For every original $k \geq 55$,

$$W_{k+1} - W_k \leq -\frac{7}{64}g(x_k)^2 - \frac{7}{128}g(x_{k-1})^2 \leq -\frac{7}{1280000}R_k^2. \tag{37}$$

Equality of successive energy values on this tail occurs exactly at the minimizer phase. Moreover,

$$\begin{aligned} W_k &\leq \rho^{k-55}W_{55}, \qquad R_k^2 \leq \frac{640000}{7}\rho^{k-55}W_{55} \quad (k \geq 55), \\ \sum_{k=55}^{\infty} R_k^2 &\leq \frac{1280000}{7}W_{55}, \qquad W_{55} \leq 2 \cdot 8^{108}R_1^2. \end{aligned} \tag{38}$$

The complete original orbit is bounded, $x_k \to x^*$, and its total increment length is finite. More precisely, for $n \geq 55$,

$$\sum_{k=n}^{\infty} |x_{k+1} - x_k| \leq \frac{1+\sqrt{\rho}}{1-\sqrt{\rho}} \sqrt{\frac{640000}{7} W_{55}}\, \rho^{(n-55)/2}. \tag{39}$$

The momentum in (34) is the bracket appearing in the ACFR energy [1, Section 3.2]; it is also the limiting normalized momentum in [3, Eqs. (4)–(6)]. Quadratic energies in positions and gradients are an established template [9, Section 4.1]. Here the two gradient-square terms retain dissipation from the corrector and control the previous position. The onset and geometric constants in the theorem are convenient bounds, with no sharpness assertion.

Proof. Strong convexity gives the unique minimizer. For every $t \in \mathbb{R}$,

$$\frac{1}{200}|t-x^*|^2 \leq f(t) - f^* \leq \frac{1}{2}|t-x^*|^2, \qquad \frac{1}{100}|t-x^*| \leq |g(t)| \leq |t-x^*|. \tag{40}$$

Write $X = x - x^*$ and $Z = z - x^*$. The integral average of f'' between x^* and z lies in $[1/100, 1]$, including when $Z = 0$. Hence

$$\left|\frac{9}{4}g(z) - Z\right| \leq \frac{5}{4}|Z|, \qquad p(x,z)^2 \leq 2X^2 + \frac{25}{8}Z^2.$$

It follows that

$$W(x,z) \leq 2X^2 + \frac{107}{64}Z^2 \leq 2R(x,z)^2, \qquad W(x,z) \geq \frac{1}{200}X^2 + \frac{7}{640000}Z^2.$$

This proves (36) on the entire phase space, before any trajectory estimate.

Each original step is explicit and globally defined. The corrector $T(t) = t - g(t)$ fixes x^* and has derivative in $[0, 99/100]$. For every $k \geq 1$, expansion of the literal predictor and (40) give

$$|y_k - x^*| \leq \frac{17}{4}|x_k - x^*| + \frac{21}{4}|x_{k-1} - x^*|.$$

Consequently

$$R_{k+1}^2 \leq \left(1 + \frac{17^2 + 21^2}{16}\right) R_k^2 = \frac{373}{8} R_k^2 \leq 64 R_k^2. \tag{41}$$

Induction yields $R_k \leq 8^{k-1} R_1$. This includes the finite negative-inertia prefix and proves the last estimate in (38) for the state actually reached at index 55.

We next use the standard smooth-convex interpolation inequality [8, Theorem 4 and Corollary 1]:

$$f(u) \geq f(v) + g(v)(u-v) + \frac{1}{2}\big(g(u) - g(v)\big)^2. \tag{42}$$

For a direct proof, the convex function $h(t) = f(t) - g(v)t$ has a minimizer at v and a 1-Lipschitz derivative. The descent estimate at $u - h'(u)$ gives $h(v) \leq h(u - h'(u)) \leq h(u) - h'(u)^2/2$, which is (42).

Fix an original index k and abbreviate

$$x = x_k, \quad z = x_{k-1}, \quad y = y_k, \quad u = x_{k+1},$$
$$a = g(x), \quad b = g(z), \quad c = g(y), \quad d = g(u).$$

With $p = p(x, z)$, define

$$q = \left(1 - \frac{4}{k}\right) p + \frac{27}{4k} b. \tag{43}$$

The original recurrence gives the exact identities

$$y = x + q - \frac{9}{4} a, \qquad u = y - c, \qquad p(u, x) = q - c. \tag{44}$$

In particular, the last term in the predictor remains $-9g(z)/(4k)$; no error sequence has been substituted for it.

Apply (42) at (y, u) and (x, y). The two defects

$$I_1 = f(y) - f(u) - \frac{1}{2}(c^2 + d^2), \qquad I_2 = f(x) - f(y) + c(y - x) - \frac{1}{2}(a - c)^2$$

are nonnegative. Expanding (34) with (44) now yields

$$W(u, x) - W(x, z) = \frac{q^2 - p^2}{2} - \frac{7}{64}(a^2 + b^2) - \frac{1}{2}\left(c + \frac{5}{4} a\right)^2 - I_1 - I_2. \tag{45}$$

The terminal gradient d cancels because its square is retained in both the corrector estimate and the energy. For $k > 2$, a second exact identity is

$$\frac{q^2 - p^2}{2} = -\frac{4(k-2)}{k^2}\left(p - \frac{27(k-4)}{32(k-2)} b\right)^2 + \frac{729}{256(k-2)} b^2. \tag{46}$$

The remaining scalar comparison is

$$\frac{7}{128} - \frac{729}{256(k-2)} = \frac{14k - 757}{256(k-2)} > 0 \qquad (k \geq 55).$$

Equations (45)–(46), followed by (40), prove (37). This is an estimate for the literal nonautonomous step, obtained before assuming any orbit bound. Equality of consecutive tail energies forces $g(x_k) = g(x_{k-1}) = 0$ and hence the minimizer phase. Conversely, that phase is fixed by every original step.

The upper bound $W_k \leq 2R_k^2$ and (37) imply $W_{k+1} \leq \rho W_k$ for $k \geq 55$. Iteration and the lower coercivity bound prove the first line of (38). Summing the stronger decrement (37) and using $W_{k+1} \geq 0$ proves its square-sum assertion. The finite original prefix is controlled by (41); the tail bounds therefore give boundedness of the complete orbit and $x_k \to x^*$.

Finally, if $A = (640000 W_{55}/7)^{1/2}$, then for $k \geq 55$,

$$|x_{k+1} - x_k| \leq R_{k+1} + R_k \leq A(1 + \sqrt{\rho})\rho^{(k-55)/2}.$$

Summation proves (39), and adding the finitely many original increments proves finite total length. If $W_{55} = 0$, the tail is fixed and all these bounds hold with $A = 0$. No division by a gradient, displacement or energy is used. ∎

The fixed potential constructed in (13) satisfies the curvature bounds in (33). Thus its original trajectories all converge at $\beta = 9/4$, whereas Theorem 1 gives a nonempty open alternating basin at $\beta = 3$. These statements concern two fixed damping choices and do not locate a transition between them.

6 Weighted energies on a minimizer interval

We keep $\alpha = 4$, $\beta = 9/4$ and $s = 1$, but now assume

$$f \in C^2(\mathbb{R}) \text{ is coercive}, \qquad 0 \le f'' \le 1, \qquad \operatorname{argmin} f = [-1, 1]. \tag{47}$$

Put $f^* = \min f$, $\varphi = f - f^*$ and $g = f'$. In particular, $\varphi(0) = g(0) = 0$. For the original index $k \ge 1$, define

$$\begin{aligned} \tau_k &= \frac{k-1}{3}, \qquad p(x,z) = x - z + \frac{9}{4} g(z), \\ U_k(x,z) &= z + \tau_k p(x,z), \\ G(x,z) &= \varphi(x) + \frac{1}{2} g(x)^2 + \frac{7}{64} g(z)^2, \\ \mathcal{E}_k^a(x,z) &= \tau_k^2 G(x,z) + \frac{1}{2}\big(U_k(x,z) - a\big)^2, \qquad a \in \{-1, 1\}. \end{aligned} \tag{48}$$

On an original trajectory, omit the arguments (x_k, x_{k-1}) and write $R_k^2 = x_k^2 + x_{k-1}^2$.

Theorem 3. *Under* (47)*, every original pair $x_0, x_1 \in \mathbb{R}$ defines a unique trajectory of* (1)*. The energies* (48) *are nonnegative and continuous. Their tail sublevels are uniformly bounded: for every $M \ge 0$ there is $B_f(M) < \infty$ such that*

$$\mathcal{E}_k^a(x,z) \le M, \quad k \ge 7, \quad a \in \{-1, 1\} \quad \Longrightarrow \quad x^2 + z^2 \le B_f(M). \tag{49}$$

For both endpoints and every original $k \ge 120$,

$$\begin{aligned} \mathcal{E}_{k+1}^a - \mathcal{E}_k^a \le & -\frac{k+1}{9}\varphi(x_k) - \frac{k^2}{576}\big(g(x_k)^2 + g(y_k)^2\big) \\ & - \frac{7(k-1)^2}{576} g(x_{k-1})^2. \end{aligned} \tag{50}$$

The complete original orbit is bounded. Both endpoint energies have finite limits, and

$$\begin{aligned} \sum_{k=120}^{\infty} (k+1)\varphi(x_k) &\le 9\mathcal{E}_{120}^a, \\ \sum_{k=120}^{\infty} k^2\big(g(x_k)^2 + g(y_k)^2\big) &\le 576\mathcal{E}_{120}^a, \\ \sum_{k=120}^{\infty} (k-1)^2 g(x_{k-1})^2 &\le \frac{576}{7}\mathcal{E}_{120}^a. \end{aligned} \tag{51}$$

There is some $\ell \in [-1,1]$ for which

$$x_k \to \ell, \qquad k(x_k - x_{k-1}) \to 0, \qquad kg(x_k) \to 0, \qquad k^2\varphi(x_k) \to 0. \tag{52}$$

The original prefix obeys

$$R_k \le 8^{k-1}R_1 \quad (k \ge 1), \qquad \mathcal{E}^a_{120} \le 1 + 12000 \cdot 8^{238}R_1^2. \tag{53}$$

No predetermined, minimum-norm or projection choice of ℓ is asserted.

The core $\tau_k^2\varphi(x_k) + (U_k - a)^2/2$ is exactly the target-dependent ACFR energy [1, Eqs. (14)–(15)]. The correction in (48) consists of the two weighted gradient squares. The use of target-dependent energies in scalar trajectory convergence also has the antecedent [12, Theorem 4.7 and Eq. (4.24)], for a different inertial recurrence. The averaging step below is elementary; see also [2, Appendix A, Lemma A.2]. The specific estimate to be proved here is (50) for the literal original IGAHD step.

Proof. Every step is globally defined. Since 0 is a minimizer and g is 1-Lipschitz,

$$0 \le \varphi(x) \le \frac{x^2}{2}, \qquad |g(x)| \le |x|, \qquad |x - g(x)| \le |x|. \tag{54}$$

All summands of (48) are nonnegative. We first verify (49) on arbitrary phases, before using a trajectory bound. Define

$$C_f(M) = \max\{|x| : \varphi(x) \le M/4\} < \infty.$$

This compact sublevel is nonempty. If $k \ge 7$ and $\mathcal{E}^a_k(x,z) \le M$, then $\tau_k \ge 2$ and

$$|x| \le C_f(M), \qquad \tau_k|g(z)| \le \frac{8}{\sqrt{7}}\sqrt{M}, \qquad |U_k| \le 1 + \sqrt{2M}.$$

The identity

$$(\tau_k - 1)z = \tau_k x + \frac{9}{4}\tau_k g(z) - U_k$$

therefore implies

$$|z| \le 2C_f(M) + \frac{18}{\sqrt{7}}\sqrt{M} + 1 + \sqrt{2M}.$$

Squaring these two coordinate bounds supplies $B_f(M)$. This also works at $M = 0$; no lower-curvature estimate is used.

The prefix estimate is equally global. By (54), expansion of the actual predictor gives

$$|y_k| \le \frac{17}{4}|x_k| + \frac{21}{4}|x_{k-1}|, \qquad R_{k+1}^2 \le \frac{373}{8}R_k^2 \le 64R_k^2.$$

This proves the first part of (53), including all negative-inertia indices. At $k = 120$, put $s = 119/3 < 40$. Since $g(z) = \theta z$ for some $\theta \in [0,1]$, including $z = 0$ by an integral average,

$$U_{120} = sx + \left(1 - s + \frac{9}{4}s\theta\right)z, \qquad U_{120}^2 \le (40^2 + 51^2)(x^2 + z^2).$$

Also $G(x,z) \le x^2 + z^2$ and $(U_{120} - a)^2/2 \le U_{120}^2 + 1$. Thus

$$\mathcal{E}_{120}^a \le 1 + 5801 R_{120}^2 \le 1 + 12000 \cdot 8^{238} R_1^2,$$

which proves the remaining prefix bound.

Fix an original index $k \ge 120$ and either endpoint a. For one actual step write

$$\begin{gathered} x = x_k, \quad z = x_{k-1}, \quad y = y_k, \quad u = x_{k+1}, \\ A = g(x), \quad b = g(z), \quad c = g(y), \quad d = g(u), \\ s = \frac{k-1}{3}, \qquad t = \frac{k}{3}. \end{gathered}$$

With $p = x - z + (9/4)b$ and $q = (1 - 4/k)p + 27b/(4k)$, direct substitution in (1) gives

$$y = x + q - \frac{9}{4}A, \qquad u = y - c, \qquad U_{k+1} = U_k - tc, \tag{55}$$

and

$$ty - (t-1)x - a = U_k - a - \frac{9}{4}tA. \tag{56}$$

These identities retain the previous-gradient correction divided by the original k.

The interpolation inequality (42) used only convexity and a 1-Lipschitz gradient, so it remains valid here. Its three defects

$$\begin{aligned} I_0 &= c(y-a) - \frac{1}{2}c^2 - \varphi(y), \\ I_1 &= \varphi(x) - \varphi(y) + c(y-x) - \frac{1}{2}(A-c)^2, \\ I_2 &= \varphi(y) - \varphi(u) - \frac{1}{2}(c^2 + d^2) \end{aligned} \tag{57}$$

are nonnegative. Here $\varphi(a) = g(a) = 0$ and $y - u = c$. Expansion with (55)–(56) yields the exact identity

$$\begin{aligned} \mathcal{E}_{k+1}^a - \mathcal{E}_k^a = &-\frac{k+1}{9}\varphi(x) - t^2 \mathcal{Q}_k(A,c) - \frac{7}{64}s^2 b^2 \\ &- tI_0 - t(t-1)I_1 - t^2 I_2, \end{aligned} \tag{58}$$

where

$$\mathcal{Q}_k(A,c) = \left(\frac{57}{64} - \frac{5}{2k} + \frac{1}{2k^2}\right)A^2 + \left(\frac{5}{4} + \frac{3}{k}\right)Ac + \frac{1}{2}c^2.$$

For example, the potential coefficient is $t(t-1) - s^2 = -(k+1)/9$. The d^2 terms cancel between the corrected potential estimate and the next gradient-square energy term. Thus (58) is a calculation for the actual weighted energies, not a multiplication of an autonomous decrement by τ_k^2.

The coefficient matrix of $\mathcal{Q}_k(A,c) - (A^2 + c^2)/64$ is

$$M_k = \begin{pmatrix} \dfrac{7}{8} - \dfrac{5}{2k} + \dfrac{1}{2k^2} & \dfrac{5}{8} + \dfrac{3}{2k} \\ \dfrac{5}{8} + \dfrac{3}{2k} & \dfrac{31}{64} \end{pmatrix}.$$

Its second diagonal entry is positive, and

$$\det M_k = \frac{17k^2 - 1580k - 1028}{512k^2}, \qquad 17k^2 - 1580k - 1028 = 17(k-120)^2 + 2500(k-120) + 54172 > 0 \tag{59}$$

for $k \geq 120$. Hence $\mathcal{Q}_k(A,c) \geq (A^2+c^2)/64$. Since $t(t-1) \geq 0$, (58) proves (50), with $t^2/64 = k^2/576$ and $7s^2/64 = 7(k-1)^2/576$.

Nonnegativity and descent give $0 \leq \mathcal{E}_k^a \leq \mathcal{E}_{120}^a$ for all $k \geq 120$. The already proved uniform sublevel bound then controls the entire tail; the prefix is controlled by (53). Both energy sequences have finite limits. Telescoping (50) proves the three assertions in (51), and also their joint bound with the coefficients in (50). In particular,

$$kg(x_k) \to 0, \qquad kg(y_k) \to 0, \qquad (k-1)g(x_{k-1}) \to 0. \tag{60}$$

The two endpoint energy limits now control motion along the interval. At every original index,

$$\mathcal{E}_k^{(-1)} - \mathcal{E}_k^{(1)} = 2U_k. \tag{61}$$

Thus $U_k \to \ell$ for some real ℓ. The actual momentum definition rearranges, for $k \geq 5$, to

$$x_k = (1-\lambda_k)x_{k-1} + \lambda_k b_k, \qquad \lambda_k = \frac{3}{k-1}, \qquad b_k = U_k - \frac{9}{4}\tau_k g(x_{k-1}) \to \ell. \tag{62}$$

Here $0 < \lambda_k < 1$ and $\sum \lambda_k = \infty$. To see convergence directly, fix $\varepsilon > 0$ and choose $N \geq 120$ with $|b_k - \ell| \leq \varepsilon$ for $k \geq N$. Iteration of (62) gives

$$|x_k - \ell| \leq P_{N,k}|x_{N-1} - \ell| + (1 - P_{N,k})\varepsilon,$$

$$P_{N,k} = \prod_{j=N}^{k} \frac{j-4}{j-1} = \frac{(N-4)(N-3)(N-2)}{(k-3)(k-2)(k-1)} \to 0.$$

Letting $k \to \infty$ and then $\varepsilon \downarrow 0$ proves $x_k \to \ell$. By (60) and continuity, $g(\ell) = 0$, so $\ell \in [-1,1]$.

Returning to the same momentum definition,

$$\tau_k(x_k - x_{k-1}) = U_k - x_{k-1} - \frac{9}{4}\tau_k g(x_{k-1}) \to 0.$$

Since $k/\tau_k \to 3$, this proves the scaled-velocity assertion without assuming it in the selection argument. Finally, the limit of either endpoint energy and (60) imply that $\tau_k^2\varphi(x_k)$ has a finite nonnegative limit. If it were positive, $(k+1)\varphi(x_k)$ would eventually dominate a positive multiple of $1/k$, contradicting (51). Hence $k^2\varphi(x_k) \to 0$.

All steps allow zero gradients, displacements and endpoint energies. An interior flat phase may move with constant endpoint energies, so strict descent on every nonstationary phase was not asserted. A single arrival position in the interval need not be stationary; the argument uses the actual two-position phase throughout. ∎

Every diagonal pair $x_0 = x_1 \in [-1,1]$ gives a stationary orbit. Thus the limit in Theorem 3 can be any point of the interval. Its scaled asymptotics do not assert finite total increment length or the geometric estimates of Theorem 2.

7 Quadratic comparison and scope

For completeness, we verify the last part of Theorem 1 directly. If $f_h(x) = \frac{h}{2}(x - x^*)^2 + f^*$ and $z_k = x_k - x^*$, the same original recurrence gives

$$z_{k+1} = (1-h)\left[\left(2 - 3h - \frac{4}{k}\right) z_k + \left(-1 + 3h + \frac{4-3h}{k}\right) z_{k-1}\right]. \tag{63}$$

This is exactly the modal recurrence in [3, Eq. (36)]. Its limiting characteristic polynomial is

$$p_h(t) = t^2 - a_h t + b_h, \qquad a_h = (1-h)(2-3h), \quad b_h = (1-h)(1-3h).$$

The real quadratic Schur conditions have the following strict margins for every $h \in [1/100, 1]$:

$$\begin{aligned} 1 - a_h + b_h &= h \geq \frac{1}{100}, \\ 1 + a_h + b_h &= 6(h - 3/4)^2 + \frac{5}{8} \geq \frac{5}{8}, \\ 1 - b_h &= h(4 - 3h) \geq \frac{397}{10000}. \end{aligned} \tag{64}$$

The last inequality follows by minimizing the concave quadratic over the stated interval. These inequalities place every limiting companion matrix strictly inside the Schur stable region.

To pass from that matrix to the actual recurrence, fix $0 < h < 1$ and let M_h be its limiting companion matrix. Its powers decay geometrically, so

$$Q_h = \sum_{j=0}^{\infty} (M_h^j)^{\mathsf{T}} M_h^j$$

is finite, $Q_h \geq I$, and $M_h^{\mathsf{T}} Q_h M_h - Q_h = -I$. Therefore M_h is strictly contractive in its Q_h norm. The actual companion matrices in (63) converge to M_h, so their norms are bounded above by some fixed number less than one from a sufficiently large original index onward. Iteration gives convergence to zero from the finite state reached at that index, for every original pair. At $h = 1$, (63) instead gives $z_k = 0$ for every $k \geq 2$. This proves the quadratic assertion without drawing an inference from a limiting matrix alone.

The normalization in [3, Proposition 6.2] is $\beta/\sqrt{s}$, with $q_{\max} = sL$. At $q_{\max} = 1$ its uniform threshold is $9/2$. Our value 3 has a strict gap of $3/2$ from that threshold. In contrast, (2) at the same point requires $\beta < 2$. The constructed potential thus shows that the entire quadratic convergence region cannot be transferred to all smooth strongly convex potentials. It does not contradict the nonlinear sufficient theorem, settle the endpoint $\beta = 2$, or preclude partial enlargements between 2 and 3. It also does not exclude other Lyapunov functions: (9) is itself a decreasing energy for the nonminimizing alternating limit.

Our comparison is with the indicated public primary versions. In [5, Theorem 4.3], the perturbation theorem concerns its specific roots-of-unity heavy-ball potential, a nearby initial pair, and uniformly small heavy-ball parameter changes and gradient errors. It permits sufficiently small $1/k$ perturbations of that system, so the stability mechanism used here is familiar. Its smooth regularization argument [5, Lemma 5.3] is also direct ancestry for preserving the local gradients in Lemma 1. The stationary framework in [6, Definition 2.1] is distinct from (1). Neither supplies the original-prefix argument of Section 4. Recent inexact IGAHD results [7] use a smaller Hessian-damping schedule and an equal-history start; their hypotheses do not include (3). These comparisons delimit the result, rather than establish an exhaustive priority claim. The theorem provides one explicit trajectory and energy obstruction to full-domain recovery, with no assertion of a sharp nonlinear parameter range.

Theorem 2 supplies a positive result at one fixed point outside the inspected nonlinear condition (2). Its momentum core comes from ACFR, its two interpolation estimates are standard, and its augmented quadratic form belongs to an established Lyapunov template [8, 9]. The specific certificate retains the terminal gradient square from the corrector and completes the actual $1/k$ momentum term before invoking boundedness. Neither eventual energy monotonicity nor attaching a finite original prefix is a new general stability principle.

The related OP26 abstract of Barik and Pattanaik [10] announces a Lyapunov analysis for a strongly convex proximal IGAHD setting. The accessible abstract supplies no recurrence, damping or step restrictions, initialization, explicit energy, or full theorem. Its overlap with Theorem 2 therefore remains undetermined. The public algorithm of He and Fang [11] uses different gradient locations and initialization, but its complete theorem text was unavailable for comparison. These limitations prevent a complete priority assertion. The results here concern the stated Lyapunov estimates and their original trajectories, without a claim of a sharp threshold or a full nonlinear convergence domain.

For Theorem 3, the announced strong-convexity scope of [10] does not contain a nontrivial minimizer interval; unannounced extensions or energy formulas in its unavailable body remain unknown. The additional certificate is the ACFR target-dependent energy augmented by the two weighted gradient squares. Polarization of its endpoint limits and the averaging argument are conventional consequences once that estimate is available. The more general displayed LT-S-IGAHD scheme of [13] also admits the algebraic substitution $\lambda_k = (9/4)(1 - 1/k)$, $\omega_k = 9/(4k)$, $\gamma_k = 0$ and $s = 1$ reproducing (1) at the present parameters. Its Theorem 20 sufficient condition then requires $(1 - 9/4)^2 < 1$, which fails. These are comparisons of the inspected statements, not exhaustive coverage assertions.

8 Local capture near a curved minimum manifold

This section changes the dimension and parameter regime. The scalar assumptions $\alpha = 4$, $s = 1$ and the preceding choices of β are no longer imposed. We consider fixed

$$d \geq 2, \qquad \alpha > 3, \qquad \beta = \sqrt{s}, \qquad 0 < sL \leq \frac{1}{4}. \tag{65}$$

Let $f : \mathbb{R}^d \to \mathbb{R}$ be C^4 and bounded below. Let M be a compact embedded C^3 submanifold without boundary, with $1 \leq \dim M \leq d-1$, consisting of local minima. For every $a \in M$, assume

$$\begin{gathered} \nabla f(a) = 0, \qquad \ker H_a = T_a M, \qquad H_a = \nabla^2 f(a), \\ \langle H_a \nu, \nu \rangle \geq \mu |\nu|^2 \quad (\nu \in N_a M), \end{gathered} \tag{66}$$

where $\mu > 0$ is fixed. Choose a tube $\mathcal{T}_\rho = \{x : \operatorname{dist}(x, M) < \rho\}$ whose closure lies inside a larger nearest-projection tube. Its nearest projection π is C^2 on a neighborhood of that closure. All derivatives used below are bounded there; let $L \geq \mu$ bound $\|\nabla^2 f\|$ on $\mathcal{T}_\rho$.

Writing $g = \nabla f$, the unchanged original recurrence is

$$\begin{aligned} y_k &= x_k + \left(1 - \frac{\alpha}{k}\right)(x_k - x_{k-1}) - s[g(x_k) - g(x_{k-1})] - \frac{s}{k} g(x_{k-1}), \\ x_{k+1} &= y_k - s g(y_k), \qquad k \geq 1. \end{aligned} \tag{67}$$

Both original positions x_0, x_1 are independent. Every predictor in (67), including the finite initial segment, will be controlled. Constants may depend on the fixed data, s and α; no uniformity as $s \downarrow 0$ or $\alpha \downarrow 3$ is asserted.

The map and its convex corrected-history energy are prior work of ACFR [2, Section 4, equations (32)–(33)]; here the source is the author manuscript dated 11 October 2021, without an identification of its body with the final published version. Attouch–Peypouquet–Redont [14] supplies the continuous DIN-AVD context, combining vanishing viscous damping with Hessian-driven damping. This attribution does not assign the first introduction of Hessian damping to that paper. The purpose below is an explicit local energy certificate and its full capture proof.

8.1 An ambient normal-history metric

All notation in this section is local. In particular, the quadratic form $\mathcal{E}_a$ and the scalar prefix constants P, F below are distinct from the earlier energies and matrix P. Write $\mathsf{T}_a, \mathsf{N}_a$ for the orthogonal tangent and normal projections, and put

$$\begin{gathered} m = s\mu, \qquad \ell = sL, \qquad Q_a = sH_a + m\mathsf{T}_a, \qquad B_a = I - Q_a, \\ \mathcal{E}_a(n, \omega) = \langle Q_a n, n \rangle + \langle B_a Q_a n, \omega \rangle + |B_a \omega|^2, \\ G_a = \begin{pmatrix} Q_a & B_a Q_a/2 \\ B_a Q_a/2 & B_a^2 \end{pmatrix}. \end{gathered} \tag{68}$$

The added $m\mathsf{T}_a$ defines an ambient metric only; it changes no tangent dynamics. For $x \in \mathcal{T}_\rho$ and $z \in \mathbb{R}^d$, set

$$\begin{gathered} a = \pi(x), \quad n = x - a, \quad v = x - z, \quad \tau = \mathsf{T}_a v, \quad \omega = \mathsf{N}_a v, \\ \mathcal{U}(x,z) = |\tau| + \sqrt{\mathcal{E}_a(n,\omega)}. \end{gathered} \tag{69}$$

Lemma 4. *The form $\mathcal{E}_a$ is positive definite on all of $\mathbb{R}^d \times \mathbb{R}^d$ and depends C^1 on a. If $S_a = \langle Q_a n, n\rangle + |B_a\omega|^2$, then*

$$\begin{gathered} \frac{3}{4} S_a \le \mathcal{E}_a(n,\omega) \le \frac{5}{4} S_a, \\ \gamma(|n|^2 + |\omega|^2) \le \mathcal{E}_a(n,\omega) \le \Gamma(|n|^2 + |\omega|^2), \qquad \gamma = \frac{3m}{4}, \quad \Gamma = \frac{5}{4}. \end{gathered} \tag{70}$$

With $c = \sqrt{3m/8}$ and $C_0 = 1 + \sqrt{5}/2$, one has

$$c\,[\operatorname{dist}(x,M) + |x - z|] \le \mathcal{U}(x,z) \le C_0[\operatorname{dist}(x,M) + |x - z|]. \tag{71}$$

The function $\mathcal{U}$ is continuous and locally Lipschitz, and its zero set is precisely $\{(a,a) : a \in M\}$. Each such state and its predictor are fixed at every original index.

Proof. The symmetric H_a vanishes on T_aM and preserves N_aM. Thus Q_a has spectrum in $[m,\ell] \subset (0, 1/4]$ and commutes with B_a. Cauchy–Schwarz gives

$$|\langle B_a Q_a n, \omega\rangle| \le \sqrt{\ell}\,\sqrt{\langle Q_a n, n\rangle}\,|B_a\omega| \le \frac{1}{4} S_a.$$

Also $Q_a \ge mI$, $B_a^2 \ge (1-\ell)^2 I \ge 9I/16 \ge mI$, and both $Q_a, B_a^2 \le I$. This proves (70). Since $|v| \le |\tau| + |\omega|$ and $|n| = \operatorname{dist}(x,M)$, its lower bound gives

$$\mathcal{U} \ge |\tau| + c(|n| + |\omega|) \ge c(|n| + |v|),$$

where $c \le 1$; the upper bound gives (71).

For clarity, the square root is handled as a norm, including at zero. Write $\|\xi\|_a = (\xi^{\mathsf{T}} G_a \xi)^{1/2}$ for ambient pairs. Compactness and the C^1 metric give, for close $a, b \in M$, a uniform L_G with $\|G_b - G_a\| \le L_G|b - a|$. Rationalizing the difference yields

$$\big|\|\xi\|_b - \|\xi\|_a\big| \le \frac{L_G}{2\sqrt{\gamma}} |b - a|\,|\xi|. \tag{72}$$

For $\xi \ne 0$, divide the difference of squared norms by their sum, which is at least $2\sqrt{\gamma}|\xi|$; for $\xi = 0$ the assertion is immediate. At fixed a the norm is $\sqrt{\Gamma}$-Lipschitz in ξ. The maps in (69) are locally Lipschitz, so these two facts prove the stated regularity. No differentiability at a norm-zero set is needed. Finally $\mathcal{U} = 0$ forces $n = \omega = \tau = 0$, hence $x = z = a \in M$; direct substitution in (67) proves fixedness. ∎

Lemma 5. *For any symmetric Q with spectrum in $[m,\ell]$, put $B=I-Q$ and define $\mathcal{E}_Q$ by* (68)*. The operators*

$$\begin{aligned} A_\infty(Q) &= \begin{pmatrix} B & B^2 \\ -Q & B^2 \end{pmatrix}, \\ A_k(Q) &= \begin{pmatrix} B(I-Q/k) & B[(1-\alpha/k)I - Q(1-1/k)] \\ B(I-Q/k)-I & B[(1-\alpha/k)I - Q(1-1/k)] \end{pmatrix} \end{aligned} \tag{73}$$

satisfy, on arbitrary ambient pairs,

$$\begin{aligned} \mathcal{E}_Q(A_\infty(n,\omega)) = \mathcal{E}_Q(Bn,B\omega), \qquad A_k = A_\infty - \mathcal{D}_Q/k, \\ \mathcal{D}_Q(n,\omega)=(r,r), \quad r = B[Qn+(\alpha I - Q)\omega], \qquad \mathcal{E}_Q(r,r)=|r|^2. \end{aligned} \tag{74}$$

Consequently, in the induced energy norm,

$$\|A_\infty\|_{\mathcal{E}_Q} \le 1-m, \qquad \|\mathcal{D}_Q\|_{\mathcal{E}_Q} \le C_D, \qquad C_D = \frac{2}{\sqrt{3}}\sqrt{\ell+\alpha^2}. \tag{75}$$

Proof. Diagonalize Q orthogonally, using the same basis in both ambient factors. For each eigenvalue q, write $b=1-q$. Direct multiplication gives

$$\begin{pmatrix} b & -q \\ b^2 & b^2 \end{pmatrix} \begin{pmatrix} q & bq/2 \\ bq/2 & b^2 \end{pmatrix} \begin{pmatrix} b & b^2 \\ -q & b^2 \end{pmatrix} = b^2 \begin{pmatrix} q & bq/2 \\ bq/2 & b^2 \end{pmatrix}.$$

Summing over the eigenbasis proves the first identity and the bound $1-m$ in every dimension. Subtracting the blocks in (73) gives $\mathcal{D}_Q$. The identity $Q+BQ+B^2=I$ gives $\mathcal{E}_Q(r,r)=|r|^2$. With $S_Q=\langle Qn,n\rangle + |B\omega|^2$,

$$|r| \le \sqrt{\ell}\,\sqrt{\langle Qn,n\rangle} + \alpha|B\omega| \le \sqrt{\ell+\alpha^2}\sqrt{S_Q} \le C_D\sqrt{\mathcal{E}_Q(n,\omega)},$$

using $\|B\|\le 1$ and $\|\alpha I - Q\| \le \alpha$. This proves the remaining bound. ∎

8.2 One-step geometry and the moving metric

Fix an integer

$$K \ge \max\left\{ \left\lceil \frac{4C_D}{m} \right\rceil, \left\lceil \frac{4\alpha}{3m} \right\rceil, \lfloor \alpha \rfloor + 1, 3 \right\}. \tag{76}$$

Lemma 6. *There exist $r_s, C_* > 0$, independent of the history and of $k \ge K$, such that whenever $x \in \mathcal{T}_\rho$ and $\mathcal{U}(x,z) \le r_s$, the actual predictor y and next point $x_+ = y - sg(y)$ in* (67) *lie in $\mathcal{T}_\rho$, as does z. Moreover,*

$$\mathcal{U}(x_+,x) \le \left(1-\frac{\alpha}{k}\right)\mathcal{U}(x,z) + C_*\mathcal{U}(x,z)^2. \tag{77}$$

Proof. Choose $0 < R < \rho/4$ small enough for the following uniform estimates. Since $D\pi(a) = \mathsf{T}_a$ on M, Taylor's formula gives

$$\begin{aligned} |g(a+e) - H_a e| \le J|e|^2, \\ |\pi(a+e) - a - \mathsf{T}_a e| \le J_\pi |e|^2 \qquad (|e| \le R). \end{aligned} \tag{78}$$

The identity for $D\pi$ follows by differentiation along tangent curves and normal lines through a. Bounded derivatives on the projection tube make J, J_π uniform. Extending the fields on M by composition with π also gives uniform constants L_T, L_G such that, for $|b-a| \le R$,

$$\|\mathsf{T}_b - \mathsf{T}_a\| \le L_T|b-a|, \qquad \|G_b - G_a\| \le L_G|b-a|. \tag{79}$$

Here the segment between the base points lies in the controlled tube. No global choice of normal frame is used.

Set $h = 1 - \alpha/k \in (0,1)$ and use the state coordinates (69). For this proof put

$$d_0 = |n| + |v|, \qquad A_y = 1 + 2\ell, \qquad A_x = (1+\ell)A_y, \qquad 0 < r_s \le \frac{cR}{2A_x}.$$

By (71), $d_0 \le r_s/c$. In particular x, z and their joining segment lie in $B(a,R)$, since $|z-a| = |n-v| \le d_0$. The actual predictor therefore satisfies

$$\begin{aligned} |y-a| &\le |n| + h|v| + \ell|v| + (\ell/k)|z-a| \le A_y d_0, \\ |x_+ - a| &\le (1+\ell)|y-a| \le A_x d_0 \le R/2. \end{aligned} \tag{80}$$

In the first line we used only the gradients at x, z, whose domains were already checked. That line places y in $B(a,R)$ before $|g(y)| \le L|y-a|$ is used in the second. Thus all four points lie in $\mathcal{T}_\rho$ without an assumption about future trapping.

Write $R_g(e) = g(a+e) - H_a e$. Expanding the literal predictor, including both old-gradient terms, gives

$$\begin{aligned} y - a &= h\tau + Y + R_y, \\ Y &= (I - Q_a/k)n + [hI - Q_a(1-1/k)]\omega, \\ R_y &= -sR_g(n) + s(1-1/k)R_g(n-v), \qquad |R_y| \le 2sJd_0^2. \end{aligned} \tag{81}$$

Both n, ω are normal at a, so Q_a acts on them as sH_a. On τ, the actual Hessian is zero. Applying the corrector and (78), now legitimately at y, gives

$$\begin{aligned} x_+ - a &= h\tau + Z + R_+, \qquad Z = B_a Y, \\ |R_+| &\le a_x d_0^2, \qquad a_x = sJ(2 + A_y^2). \end{aligned} \tag{82}$$

Indeed $\|I - sH_a\| \le 1$ and $R_+ = (I - sH_a)R_y - sR_g(y-a)$. In particular

$$(Z, Z-n) = A_k(Q_a)(n, \omega). \tag{83}$$

We next compare with the true new projection, rather than keep the old normal fiber. Let $a_+ = \pi(x_+)$, $n_+ = x_+ - a_+$, $v_+ = x_+ - x$, and let τ_+, ω_+ be its tangent and normal components at a_+. Nearest-point minimality and (80) imply

$$|a_+ - a| \le 2A_x d_0 \le R, \quad |v_+| \le (A_x+1)d_0, \quad |(n_+, \omega_+)| \le (2A_x+1)d_0.$$

The projection expansion (78) applied to $e = x_+ - a$ yields

$$\begin{aligned} |n_+ - Z| \le a_n d_0^2, \qquad |\tau_+ - h\tau| \le a_t d_0^2, \qquad |\omega_+ - (Z-n)| \le a_w d_0^2, \\ a_n = a_x + J_\pi A_x^2, \qquad a_t = a_w = a_x + 2L_T A_x(A_x+1). \end{aligned} \tag{84}$$

For the first estimate, use $n_+ = \mathsf{N}_a e - [\pi(a+e) - a - \mathsf{T}_a e]$. For the other two, first project $v_+ = h\tau + (Z-n) + R_+$ at a, then use (79). This proves that the tangent update has no linear normal-history forcing and that the normal update has no linear tangent forcing; curvature contributes only the displayed quadratic errors.

The change of metric also has quadratic size after taking the square root. Apply (72) to (n_+, ω_+) and then use (84) in the old metric:

$$\begin{aligned} \|(n_+, \omega_+)\|_{a_+} \le \|(Z, Z-n)\|_a + C_N d_0^2, \\ C_N = \sqrt{\Gamma}(a_n + a_w) + \frac{L_G A_x(2A_x+1)}{\sqrt{\gamma}}. \end{aligned} \tag{85}$$

The extra factor $|(n_+, \omega_+)|$ in (72) is essential. The ambient positive-definite extension permits this comparison even though the two pairs need not belong to the same normal fiber, and the estimate holds at zero.

Finally, Lemma 5 and (76) give

$$\|A_k(Q_a)\|_{\mathcal{E}_a} \le 1 - m + C_D/k \le 1 - 3m/4 \le h.$$

Combining (84)–(85) with $d_0 \le \mathcal{U}/c$ proves (77), for example with

$$C_* = 1 + (a_t + C_N)/c^2. \tag{86}$$

All constants were fixed from the structural data before the input state was chosen. ∎

8.3 The clock domain and the original finite prefix

Set $\eta = (\alpha - 2)/C_* > 0$ and, for original indices $k \ge K$, define

$$\begin{aligned} \mathcal{D}_k = \{(x,z) : x \in \mathcal{T}_\rho,\ \mathcal{U}(x,z) \le r_s,\ k\mathcal{U}(x,z) \le \eta\}, \\ \mathcal{W}_k(x,z) = k^2 \mathcal{U}(x,z). \end{aligned} \tag{87}$$

Along an actual trajectory abbreviate $\mathcal{U}_k = \mathcal{U}(x_k, x_{k-1})$ and $\mathcal{W}_k = k^2 \mathcal{U}_k$.

Theorem 4. *Under* (65)–(66)*, the original transition* (67) *maps* $\mathcal{D}_k$ *into* $\mathcal{D}_{k+1}$ *and satisfies*

$$\mathcal{U}_{k+1} \le (1 - 2/k)\mathcal{U}_k, \qquad \mathcal{W}_{k+1} \le \left(1 - \frac{3}{k^2} - \frac{2}{k^3}\right)\mathcal{W}_k. \tag{88}$$

Thus $\mathcal{W}$ *strictly decreases at every non-diagonal admissible transition.*

For an explicit neighborhood at the original start, choose $0 < r_{\mathrm{out}} < \rho$ *so that each* $B(a, r_{\mathrm{out}})$*,* $a \in M$*, is in the controlled tube, and set*

$$\begin{aligned} \mathsf{P} = 1 + 2(\alpha - 1) + 3sL, \qquad \mathsf{F} = \max\{1, (1+sL)\mathsf{P}\}, \\ 0 < \epsilon \le \min\{r_s, \eta/K\}, \qquad 0 < \delta \le \mathsf{F}^{-(K-1)} \min\{r_{\mathrm{out}}/4, \epsilon/9\}. \end{aligned} \tag{89}$$

Every independent original pair satisfying

$$\operatorname{dist}(x_0, M)^2 + |x_1 - x_0|^2 < \delta^2 \tag{90}$$

has all positions and all actual predictors in $\mathcal{T}_\rho$. Its state at index K belongs to $\mathcal{D}_K$, and for $k \geq K$ one has

$$\mathcal{U}_k \leq \frac{K^2 \mathcal{U}_K}{k^2}, \qquad \operatorname{dist}(x_k, M) + |x_k - x_{k-1}| \leq \frac{c^{-1} K^2 \mathcal{U}_K}{k^2}. \tag{91}$$

The actual travel $\sum_{k\geq 1} |x_k - x_{k-1}|$ is finite, and x_k converges to some $x_\infty \in M$, with the sufficient tail estimate

$$|x_k - x_\infty| \leq \frac{c^{-1} K^2 \mathcal{U}_K}{k} \qquad (k \geq K). \tag{92}$$

Proof. On $\mathcal{D}_k$, Lemma 6 first supplies an admissible next state. Since $C_* \mathcal{U}_k \leq (\alpha - 2)/k$, it gives the first inequality in (88). For $k \geq 3$ this also implies $\mathcal{U}_{k+1} \leq r_s$ and

$$(k+1)\mathcal{U}_{k+1} \leq (k+1)(1 - 2/k)\mathcal{U}_k = (k - 1 - 2/k)\mathcal{U}_k \leq \eta.$$

Hence $\mathcal{D}_k$ is mapped into $\mathcal{D}_{k+1}$. Multiplying the first inequality by $(k+1)^2$ yields the second, because

$$\frac{(k+1)^2}{k^2}\left(1 - \frac{2}{k}\right) = 1 - \frac{3}{k^2} - \frac{2}{k^3} \in (0, 1).$$

Strictness follows whenever $\mathcal{U}_k > 0$. Lemma 4 identifies the remaining states as the fixed diagonal.

It remains to reach this domain from the original two positions. Fix $a = \pi(x_0)$, which exists by (90), and put

$$d_k = \max\{|x_k - a|, |x_{k-1} - a|\}.$$

The initial condition gives $d_1 < 2\delta$. For every integer $k \geq 1$, including any negative or zero inertia coefficient, $|1 - \alpha/k| \leq \alpha - 1$. If the two current positions lie in $B(a, r_{\mathrm{out}})$, its convexity and the Hessian bound imply

$$|g(x_k) - g(x_{k-1})| \leq 2L d_k, \qquad |g(x_{k-1})| \leq L d_k,$$

and the actual predictor obeys $|y_k - a| \leq \mathsf{P} d_k$. We prove by forward induction that

$$\begin{aligned} d_k &< 2\delta \mathsf{F}^{k-1} \quad (1 \leq k \leq K), \\ |y_k - a| &\leq \mathsf{P} d_k, \qquad d_{k+1} \leq \mathsf{F} d_k \quad (k < K). \end{aligned} \tag{93}$$

Indeed $\mathsf{P} > 1$ and $\mathsf{F} = (1 + \ell)\mathsf{P} \geq \mathsf{P}$. At a step $k < K$, the inductive bound and (89) give

$$\mathsf{P} d_k < 2\delta \mathsf{P} \mathsf{F}^{k-1} \leq 2\delta \mathsf{F}^k \leq 2\delta \mathsf{F}^{K-1} \leq r_{\mathrm{out}}/2.$$

The same bounds place both current positions in the controlled ball. The predictor is therefore admissible before its gradient is estimated. Now

$$|x_{k+1} - a| \le (1 + \ell)|y_k - a| \le \mathsf{F} d_k,$$

which proves (93) and captures all prefix positions and predictors. No inverse of an early map is used.

At the terminal pair use its own nearest point $\pi(x_K)$, which need not be a. Nearest-point minimality gives $|x_K - \pi(x_K)| \le d_K$, while its velocity and each velocity projection have norm at most $2d_K$. By (70),

$$\mathcal{U}(x_K, x_{K-1}) \le 2d_K + \sqrt{\tfrac{5}{4}(d_K^2 + 4d_K^2)} = \tfrac{9}{2} d_K < 9\delta \mathsf{F}^{K-1} \le \epsilon. \tag{94}$$

Thus the actual original state enters $\mathcal{D}_K$. This is a full open two-position neighborhood of the diagonal over M, with its radius chosen before either initial position.

Starting at K, the one-step domain conclusion and the proved invariance inductively capture every remaining predictor and position. Only after this capture argument do we use $\mathcal{W}_k \le \mathcal{W}_K$ to obtain (91), the second bound following from (71). The finite prefix contributes finitely many finite increments and the tail is summable. Hence (x_k) is Cauchy, and its limit lies in the closed set M because $\operatorname{dist}(x_k, M) \to 0$. Finally

$$|x_k - x_\infty| \le \sum_{j=k+1}^{\infty} |x_j - x_{j-1}| \le c^{-1} K^2 \mathcal{U}_K \sum_{j=k+1}^{\infty} j^{-2} \le c^{-1} K^2 \mathcal{U}_K / k,$$

proving (92), also when $\mathcal{U}_K = 0$. ∎

The functions $\mathcal{U}$ and $\mathcal{W}$ are norm-based phase-distance measures on the stated domains. They are not asserted to be smooth at their norm-zero sets or Lyapunov functions outside those domains. The rates (91)–(92) are sufficient estimates only. The proof neither prescribes the limit by a projection or minimum-norm rule nor proves selector regularity or stability of an individual minimizer. Its neighborhood is at the fixed original start; it supplies no uniform unscaled radius for arbitrary late entry times or conclusion for general β.

8.4 Scope of the source comparison

Nearby-history capture and finite travel around Morse–Bott minima already occur for fixed-momentum NAG in Feng–Yuan [15, Theorems 2.1–2.2 and Sections 3.2–3.3]. That comparison includes genuine trapping from nearby independent positions, rather than just a rate conditional on convergence. The time-varying map (67) and the explicit metric (68) are the objects treated here.

General invariant-manifold theory is also relevant. Bento–Silva [16, Theorem 2 and Examples 7–8] permits nonuniform polynomial stable graphs and is a plausible route to

a qualitative result. The vanishing tangent damping alone does not exclude such a route. A verified application would still need the actual normal and nonlinear estimates, as well as a parameterized family whose union covers the full nearby two-position space; a stable graph at one prescribed reference point does not by itself supply that coverage. No obstruction or novelty is inferred from leaving that application open. The present proof instead gives the ambient metric identity, the moving-metric estimate, the strict clock-domain descent and the explicit original-prefix radius directly. The source comparison is limited to the indicated inspected versions and is not a comprehensive comparison of generic foliation results or a priority claim.

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